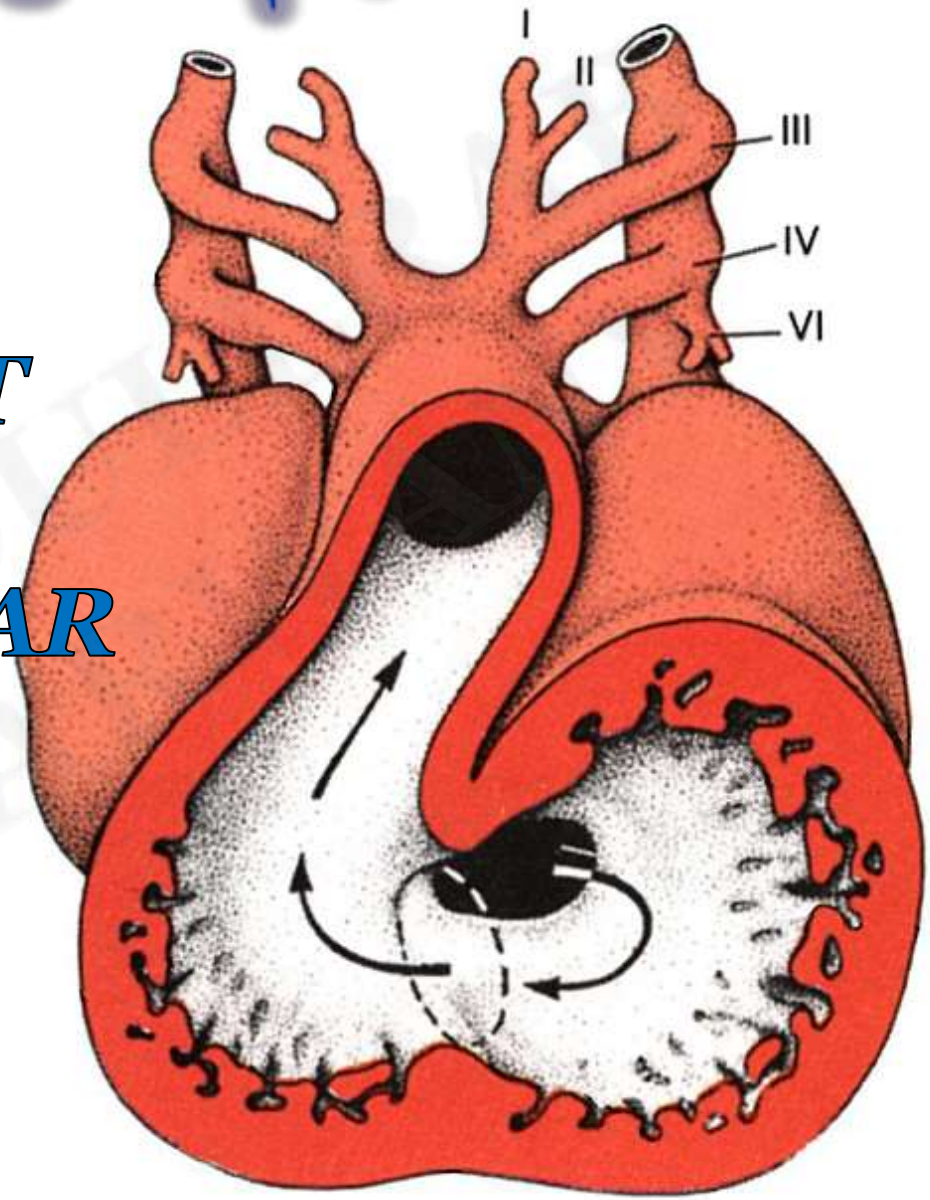
بسم الله الرحمن الرحيم

DEVELOPMENT OF CARDIOVASCULAR SYSTEM



DR. ARAB AL-MARWANY

INTRODUCTION

*The nutritional supply of embryo **changes** according to development as follows:*

- ***During the first week**, before implantation, **deutoplasm** (accumulated cytoplasmic nutrients) of oocyte supplies nutrition.*
- ***During the second week**, breakdown products of **endometrium** (due to implantation) nourishes the embryo by simple diffusion.*
- ***After the third week**, **maternal blood** nourish embryo through uteroplacental circulation.*

- *Increasing nutritional demand of growing embryo initiates development of cardiovascular system in the third week of intrauterine life.*

- *All components of cardiovascular system develop from **Mesoderm**.*
- *The components of cardiovascular system can be studied as*

1. Development of heart

2. Development of blood vessels

DR. ARAB AL-MARWANY

Development Of The Heart

DR ABDULLRAB AL MARWANY

DR. A.RAB AL - MARWANY

Lets Remember The Result Of The Head Folding &Intra-embryonic coelom:

Intra-embryonic coelom:

Definition:

It is a horse shoe cavity that is formed in the lateral plate mesoderm.

- *Its caudal end **communicate** with the extra-embryonic coelom at the margin of embryonic disc.*

- *It is **divided into**:*

***a) Pericardial cavity** in median cranial part of the coelom.*

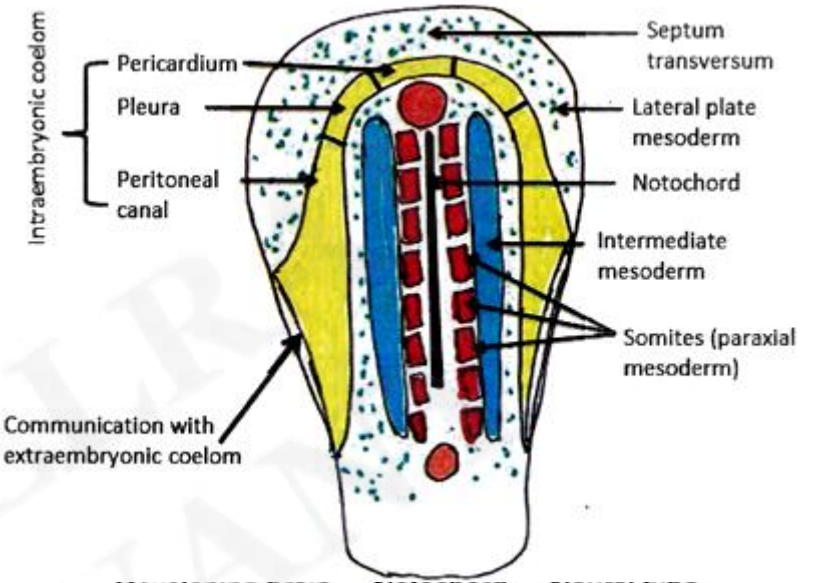
***b) Peritoneal canals** in the lateral part of the coelom.*

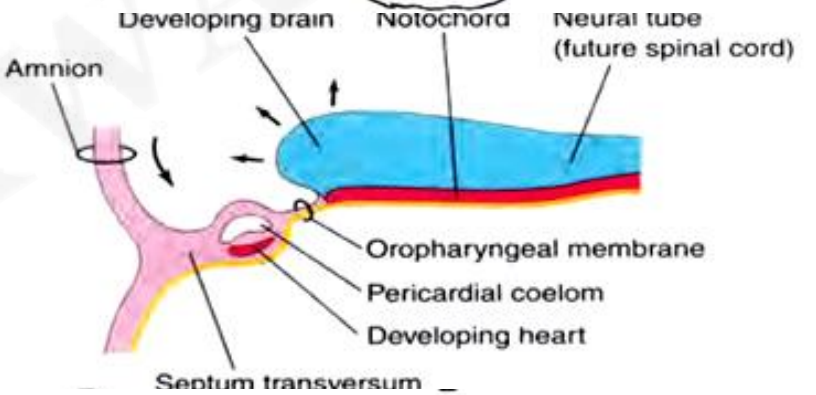
***c) Pleural cavities** between pericardium and peritoneal canals.*

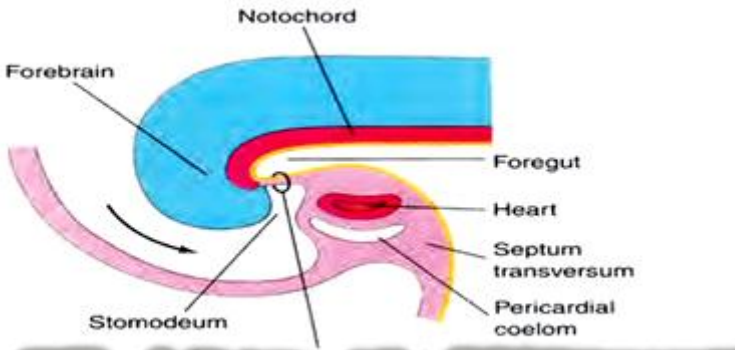
N.B.

Cardiogenic area is the splanchnic mesoderm related to pericardium.

*Septum transversum is a mass of mesoderm **cranial** to pericardium (**before folding**) and **caudal** to pericardium (**after folding**).*







DR. ARAB AL-MARWANY